

ESSIAW CHARLES JNR

Software Developer & Full-Stack Systems Engineer

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[Portfolio](#) | [GitHub](#) | [LinkedIn](#)

PROFESSIONAL SUMMARY

Full-stack software developer specializing in resilient web applications, real-time event-driven systems, and clean UI engineering. Experience architecting production platforms including a cloud restaurant POS/KDS system, an emergency dispatch coordination platform, and a cryptographic QR access-control system. Proficient in TypeScript, Next.js (App Router), Supabase, PostgreSQL, Python, and Docker, with a consistent focus on sub-second latency, strong Core Web Vitals scores, and modular, testable code.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), Python, SQL (PostgreSQL), HTML5, CSS3, Bash

Frontend:

KEY PROJECTS

Sweetbite Food POS & Kitchen Display System (KDS)

Active Production System

Next.js 14 · TypeScript · Supabase Realtime · PostgreSQL · Tailwind CSS

- Built an end-to-end cloud POS and Kitchen Display System for food service operations, replacing paper tickets and reducing kitchen communication delays.
- Used Supabase Realtime WebSocket channels with bi-directional state sync to broadcast order and ticket status updates across kitchen stations in under 500ms.
- Designed kitchen alert chimes, high-contrast touch ticket layouts, automated table routing, and optimistic UI updates with offline/reconnection recovery.
- Modeled normalized PostgreSQL schemas for order states, line items, and station routing to prevent order collisions during peak hours.

Ghana Emergency Response & Dispatch Telemetry System

Live Deployment

TypeScript · JavaScript · Python · PostgreSQL · Geolocation API · WebRTC

- Built an emergency web application for geolocation-tagged SOS requests to support rapid dispatch coordination across Ghana.
- Implemented live coordinate telemetry with interactive mapping, geofencing, and voice-note transmission to speed up incident verification.
- Added client-side offline queues and service-worker persistence to preserve distress requests during intermittent connectivity.
- Built an administrative dispatch console for triaging incidents by severity, viewing nearby units, and updating live call status.

Patron Housing Access Control & Visitor Management

Production Solution

Python · PostgreSQL · Docker · Cryptographic QR Codes · Modern CSS

- Built a residential access-control platform for digital visitor passes, gate check-ins, and cryptographic QR code validation.
- Created an administrative dashboard with real-time entry/exit logs and resident authorization workflows; deployed with Docker on Render.
- Implemented automated pass revocation and single-use visitor passes to prevent credential sharing.

CLIENT & COMMERCIAL PLATFORMS

BizConnect Technologies — Enterprise Portal

Client Production

Modern JavaScript · Accessible UI/UX · REST APIs

- Built a corporate web platform with a lean codebase, responsive layouts, and a 100% Core Web Vitals score.
- Implemented WCAG 2.1 AA accessible typography, keyboard navigation, and semantic HTML structure.
- Built a modular component system and contact-inquiry API integration that drove a 35% increase in verified inbound leads.

Kelrose Tours — Travel & Itinerary Booking Platform

Client Production

JavaScript · Responsive CSS · SEO

- Designed and deployed a travel platform with interactive tour itinerary catalogs and destination guides.
- Optimized responsive layout, image delivery, and metadata indexing to improve search visibility and mobile retention.
- Configured automated inquiry workflows routing customer booking requests to operations staff.

PROFESSIONAL EXPERIENCE

Full-Stack Systems Engineer & Independent Developer

2024 – Present

Independent / Client Engagements — Accra, Ghana

- Lead full-lifecycle engineering for web systems across hospitality, municipal emergency, and commercial sectors.
- Design PostgreSQL database models, implement role-based access control, and build asynchronous, event-driven services deployed on cloud infrastructure.
- Maintain a 5.0 client satisfaction rating (Google Reviews) through careful requirement scoping and disciplined delivery.

Commercial Web Engineer & Client Consultant

2023 – 2024

Freelance & Contract Engagements

- Modernized legacy web applications for SME clients, migrating static pages to interactive stacks and improving conversion funnels.
- Ran performance audits and asset optimization that cut page load times by over 40%, achieving sub-second Largest Contentful Paint.
- Delivered responsive, mobile-first UI components tested across devices and legacy browsers.

Software Engineering Apprenticeship & Foundations

2022 – 2023

Self-Directed Study — 1,500+ hours

- Completed a self-directed curriculum covering algorithms, asynchronous concurrency, network protocols (HTTP/REST/WebSockets), and relational database design.
- Adopted AI-assisted development workflows to prototype, refactor, and test code more efficiently.
- Built and shared open-source web components with local developer communities in Accra.

EDUCATION & CONTINUOUS LEARNING

Software Engineering & Systems Development — Accra, Ghana

Self-directed curricula, open-source contributions, and professional practice

- Ongoing study in Next.js App Router, advanced TypeScript, PostgreSQL query optimization, and cloud-native deployment.